Derivation of elephant induced pluripotent stem cells

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Abstract

Biodiversity loss in the present era requires new tools for studying nonmodel organisms. Elephants are both an endangered species and excellent models for studying complex phenotypes including size, social behavior and longevity. Here we report the first derivation of elephant (*Elephas maximus*) induced pluripotent stem (emiPS) cells. We achieved emiPS cells using two approaches: (1) a two-step process of chemical media induction and colony selection followed by over-expression of elephant transcription factors; and (2) a one-step process with transcription factors and *HRAS* mutant, $HRAS_{G12V}$. For both protocols, we inhibited TP53 retrogenes, which are hypothesized to confer unique cancer resistance in elephants. To confirm their reprogrammed state, we generated a functional omics catalog of emiPS cells. While these emiPS cells remain transgene-dependent, we inactivated the transgenes and differentiated emiPS cells into all three germ layers via tri-lineage differentiation, embryoid body generation and direct differentiation into putative cell types from all three layers. These methods will open new frontiers for cellular models of nonmodel organisms, including for genetic rescue and conservation.